Newman-Penrose Formalism Calculations for

metric:

Name: Stephani et al., Equation (22.40)

Reference(s): [1]

Line element:

$$ds^2 = -e^{(2\,k-2\,U(t,\rho))}\mathrm{d}t \otimes \mathrm{d}t + e^{(2\,k-2\,U(t,\rho))}\mathrm{d}\rho \otimes \mathrm{d}\rho + e^{(2\,U(t,\rho))}\mathrm{d}z \otimes \mathrm{d}z + A\left(t,\rho\right)e^{(2\,U(t,\rho))}\mathrm{d}z \otimes \mathrm{d}\phi + A\left(t,\rho\right)e^{(2\,U(t,\rho))}\mathrm{d}\phi \otimes \mathrm{d}z + \left(A\left(t,\rho\right)^2 e^{(4\,U(t,\rho))} + W\left(t,\rho\right)^2\right)e^{(-2\,U(t,\rho))}\mathrm{d}\phi \otimes \mathrm{d}\phi \tag{1}$$

Coordinates: (t, ρ, z, ϕ)

Null tetrad components:

Covariant components:

$$l = \left(\frac{1}{2}\sqrt{2}e^{(k-U(t,\rho))}\right)dt + \left(-\frac{1}{2}\sqrt{2}e^{(k-U(t,\rho))}\right)d\rho + (0)\,dz + (0)\,d\phi \tag{2}$$

$$n = \left(\frac{1}{2}\sqrt{2}e^{(k-U(t,\rho))}\right)dt + \left(\frac{1}{2}\sqrt{2}e^{(k-U(t,\rho))}\right)d\rho + (0)\,dz + (0)\,d\phi \tag{3}$$

$$m = (0)\,dt + (0)\,d\rho + \left(\frac{1}{2}\sqrt{2}e^{(U(t,\rho))}\right)dz + \left(\frac{1}{2}\sqrt{2}\left(i\,W\left(t,\rho\right)e^{(-U(t,\rho))} + A\left(t,\rho\right)e^{(U(t,\rho))}\right)\right)d\phi \tag{4}$$

$$\overline{m} = (0)\,dt + (0)\,d\rho + \left(\frac{1}{2}\sqrt{2}e^{(U(t,\rho))}\right)dz + \left(\frac{1}{2}\sqrt{2}\left(-i\,W\left(t,\rho\right)e^{(-U(t,\rho))} + A\left(t,\rho\right)e^{(U(t,\rho))}\right)\right)d\phi \tag{5}$$

Contravariant components:

$$l = \left(-\frac{1}{2}\sqrt{2}e^{(-k+U(t,\rho))}\right) dt + \left(-\frac{1}{2}\sqrt{2}e^{(-k+U(t,\rho))}\right) d\rho + (0)\, dz + (0)\, d\phi \tag{6}$$

$$n = \left(-\frac{1}{2}\sqrt{2}e^{(-k+U(t,\rho))}\right) dt + \left(\frac{1}{2}\sqrt{2}e^{(-k+U(t,\rho))}\right) d\rho + (0)\, dz + (0)\, d\phi \tag{7}$$

$$m = (0)\, dt + (0)\, d\rho + \left(\frac{\left(-i\sqrt{2}A(t,\rho)\, e^{(2\,U(t,\rho))} + \sqrt{2}W(t,\rho)\right)e^{(-U(t,\rho))}}{2\,W(t,\rho)}\right) dz + \left(\frac{i\sqrt{2}e^{(U(t,\rho))}}{2\,W(t,\rho)}\right) d\phi \tag{8}$$

$$\overline{m} = (0)\, dt + (0)\, d\rho + \left(\frac{\left(i\sqrt{2}A(t,\rho)\, e^{(2\,U(t,\rho))} + \sqrt{2}W(t,\rho)\right)e^{(-U(t,\rho))}}{2\,W(t,\rho)}\right) dz + \left(-\frac{i\sqrt{2}e^{(U(t,\rho))}}{2\,W(t,\rho)}\right) d\phi \tag{9}$$

Directional derivatives:

$$D = -\frac{1}{2}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}f\left(t,\rho,z,\phi\right) - \frac{1}{2}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}f\left(t,\rho,z,\phi\right) \tag{10}$$

$$\Delta = -\frac{1}{2}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}f\left(t,\rho,z,\phi\right) + \frac{1}{2}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}f\left(t,\rho,z,\phi\right) \tag{11}$$

$$\delta = \frac{1}{2}\sqrt{2}e^{(-U(t,\rho))}\frac{\partial}{\partial z}f\left(t,\rho,z,\phi\right) - \frac{i\sqrt{2}A\left(t,\rho\right)e^{(U(t,\rho))}\frac{\partial}{\partial z}f\left(t,\rho,z,\phi\right)}{2\,W\left(t,\rho\right)} + \frac{i\sqrt{2}e^{(U(t,\rho))}\frac{\partial}{\partial \phi}f\left(t,\rho,z,\phi\right)}{2\,W\left(t,\rho\right)} \tag{12}$$

$$\overline{\delta} = \frac{1}{2}\sqrt{2}e^{(-U(t,\rho))}\frac{\partial}{\partial z}f\left(t,\rho,z,\phi\right) + \frac{i\sqrt{2}A\left(t,\rho\right)e^{(U(t,\rho))}\frac{\partial}{\partial z}f\left(t,\rho,z,\phi\right)}{2\,W\left(t,\rho\right)} - \frac{i\sqrt{2}e^{(U(t,\rho))}\frac{\partial}{\partial \phi}f\left(t,\rho,z,\phi\right)}{2\,W\left(t,\rho\right)} \tag{13}$$

Spin coefficients:

$$\rho = \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} + \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)} \tag{14}$$

$$\begin{aligned}\sigma = {} & \frac{1}{2}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{2}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{4\,W(t,\rho)} + \frac{i\sqrt{2}e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{4\,W(t,\rho)} \\ & - \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} - \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)}\end{aligned} \tag{15}$$

$$\kappa = 0 \tag{16}$$

$$\mu = -\frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} + \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)} \tag{17}$$

$$\lambda = \tag{18}$$

$$-\frac{\left(2\sqrt{2}W(t,\rho)\,e^{(U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) - 2\sqrt{2}W(t,\rho)\,e^{(U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) - i\sqrt{2}e^{(3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho) + i\sqrt{2}e^{(3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho) - \sqrt{2}e^{(U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho) + \sqrt{2}e^{(U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)\right)e^{(}}{4\,W(t,\rho)}$$

$$\nu = 0 \tag{19}$$

$$\alpha = 0 \tag{20}$$

$$\beta = 0 \tag{21}$$

$$\tau = 0 \tag{22}$$

$$\pi = 0 \tag{23}$$

$$\epsilon = \frac{1}{4}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{4}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{8\,W(t,\rho)} + \frac{i\sqrt{2}e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{8\,W(t,\rho)} \tag{24}$$

$$\gamma = -\frac{1}{4}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{4}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{8\,W(t,\rho)} - \frac{i\sqrt{2}e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{8\,W(t,\rho)} \tag{25}$$

Relations between spin coefficients:

Ricci tensor components:

$$\Phi_{00} = -\frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)^{2} - e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right) - \frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U\left(t,\rho\right)^{2} - \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)^{2}}{8\,W\left(t,\rho\right)^{2}}$$
$$- \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)^{2}} - \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)^{2}}{8\,W\left(t,\rho\right)^{2}} - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^{2}}{(\partial t)^{2}}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^{2}}{\partial t\partial \rho}W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^{2}}{(\partial \rho)^{2}}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \tag{26}$$

$$\Phi_{01} = 0 \tag{27}$$

$$\Phi_{02} = \frac{i\, e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial t}U\left(t,\rho\right)}{W\left(t,\rho\right)} + \frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^{2}}{(\partial t)^{2}}U\left(t,\rho\right) - \frac{i\, e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right)}{W\left(t,\rho\right)} - \frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^{2}}{(\partial \rho)^{2}}U\left(t,\rho\right)$$
$$+ \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)\frac{\partial}{\partial t}W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U\left(t,\rho\right)\frac{\partial}{\partial \rho}W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)^{2}}{4\,W\left(t,\rho\right)^{2}} + \frac{i\, e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^{2}}{(\partial t)^{2}}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)}$$
$$+ \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)^{2}}{4\,W\left(t,\rho\right)^{2}} - \frac{i\, e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^{2}}{(\partial \rho)^{2}}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{i\, e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial t}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^{2}} - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^{2}}{(\partial t)^{2}}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)}$$
$$+ \frac{i\, e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial \rho}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^{2}} + \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^{2}}{(\partial \rho)^{2}}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \tag{28}$$

$$\Phi_{10} = 0 \tag{29}$$

$$\Phi_{11} = -\frac{1}{4}\, e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)^{2} + \frac{1}{4}\, e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U\left(t,\rho\right)^{2} + \frac{1}{4}\, e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^{2}}{(\partial t)^{2}}U\left(t,\rho\right) - \frac{1}{4}\, e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^{2}}{(\partial \rho)^{2}}U\left(t,\rho\right)$$
$$+ \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)\frac{\partial}{\partial t}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U\left(t,\rho\right)\frac{\partial}{\partial \rho}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)^{2}}{16\,W\left(t,\rho\right)^{2}} + \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)^{2}}{16\,W\left(t,\rho\right)^{2}} \tag{30}$$

$$\Phi_{12} = 0 \tag{31}$$

$$\begin{aligned}\Phi_{20} = &-\frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)\,\frac{\partial}{\partial t}U(t,\rho)}{W(t,\rho)} + \frac{1}{2}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}U(t,\rho) + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)\,\frac{\partial}{\partial \rho}U(t,\rho)}{W(t,\rho)} - \frac{1}{2}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}U(t,\rho)\\
&+\frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho)\,\frac{\partial}{\partial t}W(t,\rho)}{2\,W(t,\rho)} - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho)\,\frac{\partial}{\partial \rho}W(t,\rho)}{2\,W(t,\rho)} - \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)^2}{4\,W(t,\rho)^2} - \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}A(t,\rho)}{4\,W(t,\rho)}\\
&+\frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)^2}{4\,W(t,\rho)^2} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}A(t,\rho)}{4\,W(t,\rho)} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)\,\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)^2} - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}W(t,\rho)}{4\,W(t,\rho)}\\
&-\frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)\,\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}W(t,\rho)}{4\,W(t,\rho)}\end{aligned} \tag{32}$$

$$\Phi_{21} = 0 \tag{33}$$

$$\begin{aligned}\Phi_{22} = &-\frac{1}{2}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho)^2 + e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho)\,\frac{\partial}{\partial \rho}U(t,\rho) - \frac{1}{2}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho)^2 - \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)^2}{8\,W(t,\rho)^2}\\
&+\frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)\,\frac{\partial}{\partial \rho}A(t,\rho)}{4\,W(t,\rho)^2} - \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)^2}{8\,W(t,\rho)^2} - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}W(t,\rho)}{4\,W(t,\rho)} + \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}W(t,\rho)}{2\,W(t,\rho)} - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}W(t,\rho)}{4\,W(t,\rho)}\end{aligned} \tag{34}$$

$$\begin{aligned}\Lambda = &\frac{1}{12}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho)^2 - \frac{1}{12}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho)^2 - \frac{1}{12}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}U(t,\rho)\\
&+\frac{1}{12}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}U(t,\rho) - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho)\,\frac{\partial}{\partial t}W(t,\rho)}{12\,W(t,\rho)} + \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho)\,\frac{\partial}{\partial \rho}W(t,\rho)}{12\,W(t,\rho)}\\
&+\frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)^2}{48\,W(t,\rho)^2} - \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)^2}{48\,W(t,\rho)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}W(t,\rho)}{12\,W(t,\rho)} - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}W(t,\rho)}{12\,W(t,\rho)}\end{aligned} \tag{35}$$

Weyl tensor components:

$$
\begin{aligned}
\Psi_0 = & -e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)^2 - 2\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right) - e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U\left(t,\rho\right)^2 - \frac{3i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial t}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& - \frac{3i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial t}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{1}{2}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}U\left(t,\rho\right) - e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}U\left(t,\rho\right) - \frac{3i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& - \frac{3i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{1}{2}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}U\left(t,\rho\right) + \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} - \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\
& - \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}A\left(t,\rho\right)}{2\,W\left(t,\rho\right)} + \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}A\left(t,\rho\right)}{2\,W\left(t,\rho\right)^2} + \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} - \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\
& + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial t}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial t}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial \rho}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)}
\end{aligned}
\tag{36}
$$

$$
\Psi_1 = 0 \tag{37}
$$

$$
\begin{aligned}
\Psi_2 = & \frac{1}{3}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)^2 - \frac{1}{3}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U\left(t,\rho\right)^2 + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial t}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} + \frac{1}{6}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}U\left(t,\rho\right) \\
& - \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{1}{6}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}U\left(t,\rho\right) - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)\frac{\partial}{\partial t}W\left(t,\rho\right)}{3\,W\left(t,\rho\right)} + \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U\left(t,\rho\right)\frac{\partial}{\partial \rho}W\left(t,\rho\right)}{3\,W\left(t,\rho\right)} \\
& + \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)^2}{12\,W\left(t,\rho\right)^2} - \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)^2}{12\,W\left(t,\rho\right)^2} - \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial t}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}W\left(t,\rho\right)}{12\,W\left(t,\rho\right)} \\
& + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}W\left(t,\rho\right)}{12\,W\left(t,\rho\right)}
\end{aligned}
\tag{38}
$$

$$\Psi_3 = 0 \tag{39}$$

$$\begin{aligned}
\Psi_4 = & -e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)^2 + 2\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right) - e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U\left(t,\rho\right)^2 + \frac{3i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial t}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& - \frac{3i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial t}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{1}{2}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}U\left(t,\rho\right) + e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}U\left(t,\rho\right) - \frac{3i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& + \frac{3i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{1}{2}\,e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}U\left(t,\rho\right) + \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\
& - \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}A\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}A\left(t,\rho\right)}{2\,W\left(t,\rho\right)^2} + \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\
& - \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial t}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial t}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial \rho}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)}
\end{aligned} \tag{40}$$

Newman-Penrose equations:

$$NP1 = 0 \tag{41}$$

$$NP2 = 0 \tag{42}$$

$$NP3 = 0 \tag{43}$$

$$NP4 = 0 \tag{44}$$

$$NP5 = 0 \tag{45}$$

$$NP6 = 0 \tag{46}$$

$$NP7 = 0 \tag{47}$$

$$NP8 = 0 \tag{48}$$

$$NP9 = 0 \tag{49}$$

$$NP10 = 0 \tag{50}$$

$$NP11 = 0 \tag{51}$$

$$NP12 = 0 \tag{52}$$

$$NP13 = 0 \tag{53}$$

$$NP14 = 0 \tag{54}$$

$$NP15 = 0 \tag{55}$$

$$NP16 = 0 \tag{56}$$

$$NP17 = 0 \tag{57}$$

$$NP18 = 0 \tag{58}$$

Bianchi identities:

$$BI1 = 0 \tag{59}$$

$$BI2 = 0 \tag{60}$$

$$BI3 = 0 \tag{61}$$

$$BI4 = 0 \tag{62}$$

$$BI5 = 0 \tag{63}$$

$$BI6 = 0 \tag{64}$$

$$BI7 = 0 \tag{65}$$

$$BI8 = 0 \tag{66}$$

$$BI9 = 0 \tag{67}$$

$$BI10 = 0 \tag{68}$$

$$BI11 = 0 \tag{69}$$

Petrov classification:

Petrov class:

A Type transformations:

Spin coefficients:

$$\rho' = \rho \tag{70}$$

$$\sigma' = \sigma \tag{71}$$

$$\kappa' = 0 \tag{72}$$

$$\begin{aligned}\mu' = {} & \sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}z^2 e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{2\,W(t,\rho)} + \frac{i\sqrt{2}z^2 e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{2\,W(t,\rho)} - \frac{\sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} \\ & - \frac{\sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)} + \frac{\sqrt{2}z e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{2\,W(t,\rho)} + \frac{\sqrt{2}z e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{2\,W(t,\rho)} - \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} + \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)}\end{aligned} \tag{73}$$

$$\lambda' = \lambda \tag{74}$$

$$\begin{aligned}\nu' = {} & \frac{1}{2}\sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{2}\sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}z^2 e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{4\,W(t,\rho)} \\ & + \frac{i\sqrt{2}z^2 e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{4\,W(t,\rho)} + \frac{\sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} + \frac{\sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)}\end{aligned} \tag{75}$$

$$\begin{aligned}\alpha' = {} & \frac{1}{4}\sqrt{2}z e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{4}\sqrt{2}z e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}z e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{8\,W(t,\rho)} \\ & + \frac{i\sqrt{2}z e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{8\,W(t,\rho)} + \frac{\sqrt{2}z e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} + \frac{\sqrt{2}z e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)}\end{aligned} \tag{76}$$

$$\beta' = \frac{3}{4}\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{3}{4}\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{3i\sqrt{2}ze^{(-k+3U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{8W(t,\rho)} + \frac{3i\sqrt{2}ze^{(-k+3U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{8W(t,\rho)} - \frac{\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4W(t,\rho)} - \frac{\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4W(t,\rho)} \tag{77}$$

$$\tau' = \frac{1}{4}\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{4}\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}ze^{(-k+3U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{8W(t,\rho)} + \frac{i\sqrt{2}ze^{(-k+3U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{8W(t,\rho)} \tag{78}$$

$$\pi' = \frac{1}{2}\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{2}\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}ze^{(-k+3U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{4W(t,\rho)} + \frac{i\sqrt{2}ze^{(-k+3U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{4W(t,\rho)} - \frac{\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4W(t,\rho)} - \frac{\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4W(t,\rho)} \tag{79}$$

$$\epsilon' = \epsilon \tag{80}$$

$$\begin{aligned}\gamma' = {} & \frac{3}{4}\sqrt{2}z^2e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{3}{4}\sqrt{2}z^2e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{3i\sqrt{2}z^2e^{(-k+3U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{8W(t,\rho)} + \frac{3i\sqrt{2}z^2e^{(-k+3U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{8W(t,\rho)} \\ & + \frac{1}{4}\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{4}\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) - \frac{\sqrt{2}z^2e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4W(t,\rho)} - \frac{\sqrt{2}z^2e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4W(t,\rho)} \\ & + \frac{i\sqrt{2}ze^{(-k+3U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{8W(t,\rho)} + \frac{i\sqrt{2}ze^{(-k+3U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{8W(t,\rho)} - \frac{1}{4}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{4}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) \\ & + \frac{\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4W(t,\rho)} + \frac{\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4W(t,\rho)} + \frac{i\sqrt{2}e^{(-k+3U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{8W(t,\rho)} - \frac{i\sqrt{2}e^{(-k+3U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{8W(t,\rho)}\end{aligned} \tag{81}$$

Weyl components:

$$\Psi_0' = \Psi_0 \tag{82}$$

$$
\begin{aligned}
\Psi_1' = & -ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)^2 - 2\,ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right) - ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U\left(t,\rho\right)^2 - \frac{3i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial t}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& - \frac{3i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial t}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{1}{2}\,ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}U\left(t,\rho\right) - ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}U\left(t,\rho\right) - \frac{3i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& - \frac{3i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{1}{2}\,ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}U\left(t,\rho\right) + \frac{ze^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} - \frac{i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\
& - \frac{i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}A\left(t,\rho\right)}{2\,W\left(t,\rho\right)} + \frac{ze^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}A\left(t,\rho\right)}{2\,W\left(t,\rho\right)^2} + \frac{ze^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} - \frac{i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\
& + \frac{i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial t}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial t}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& + \frac{i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial \rho}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)}
\end{aligned}
\tag{83}
$$

$$
\begin{aligned}
\Psi_2' = & -z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - 2\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 - \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} \\
& - \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) - z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} U(t,\rho) \\
& - \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} - \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) \\
& + \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{4\,W(t,\rho)^2} - \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} A(t,\rho)}{4\,W(t,\rho)} - \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} A(t,\rho)}{2\,W(t,\rho)} + \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{2\,W(t,\rho)^2} \\
& + \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{4\,W(t,\rho)^2} - \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} A(t,\rho)}{4\,W(t,\rho)} + \frac{1}{3}\,e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - \frac{1}{3}\,e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 \\
& + \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\,W(t,\rho)} \\
& + \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)} \\
& + \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} + \frac{1}{6}\,e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) - \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{6}\,e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) \\
& - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{3\,W(t,\rho)} + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{3\,W(t,\rho)} + \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{12\,W(t,\rho)^2} - \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{12\,W(t,\rho)^2} \\
& - \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{12\,W(t,\rho)} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{12\,W(t,\rho)}
\end{aligned}
\tag{84}
$$

$$
\begin{aligned}
\Psi_3' = & -z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - 2\,z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 - \frac{3i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} \\
& - \frac{3i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) - z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} U(t,\rho) \\
& - \frac{3i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} - \frac{3i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) \\
& + \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{4\,W(t,\rho)^2} - \frac{i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} A(t,\rho)}{4\,W(t,\rho)} - \frac{i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} A(t,\rho)}{2\,W(t,\rho)} + \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{2\,W(t,\rho)^2} \\
& + \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{4\,W(t,\rho)^2} - \frac{i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} A(t,\rho)}{4\,W(t,\rho)} + z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 \\
& + \frac{i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} + \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\,W(t,\rho)} \\
& + \frac{i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)} \\
& + \frac{3i\,z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} + \frac{1}{2}\,z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) - \frac{3i\,z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) \\
& - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{W(t,\rho)} + \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{W(t,\rho)} + \frac{z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{4\,W(t,\rho)^2} - \frac{z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{4\,W(t,\rho)^2} \\
& - \frac{3i\,z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} + \frac{3i\,z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)}
\end{aligned}
\tag{85}
$$

$$
\begin{aligned}
\Psi_4' = & -z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - 2\,z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 - \frac{3i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} \\
& - \frac{3i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) - z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} U(t,\rho) - \frac{3i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} \\
& - \frac{3i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) + \frac{z^4 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{4\,W(t,\rho)^2} - \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} A(t,\rho)}{4\,W(t,\rho)} \\
& - \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} A(t,\rho)}{2\,W(t,\rho)} + \frac{z^4 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{2\,W(t,\rho)^2} + \frac{z^4 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{4\,W(t,\rho)^2} - \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} A(t,\rho)}{4\,W(t,\rho)} \\
& + 2\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - 2\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 + \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} \\
& + \frac{z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} + \frac{z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\,W(t,\rho)} + \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} \\
& + \frac{z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)} + \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{W(t,\rho)} + z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) - \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{W(t,\rho)} \\
& - z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) - \frac{2\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{W(t,\rho)} + \frac{2\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{W(t,\rho)} + \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{2\,W(t,\rho)^2} \\
& - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{2\,W(t,\rho)^2} - e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 + 2\,e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 \\
& - \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{2\,W(t,\rho)^2} + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{2\,W(t,\rho)} + \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{2\,W(t,\rho)^2} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{2\,W(t,\rho)} \\
& + \frac{3i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} - \frac{3i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) + e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} U(t,\rho) \\
& - \frac{3i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} + \frac{3i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) + \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{4\,W(t,\rho)^2} \\
& + \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} A(t,\rho)}{4\,W(t,\rho)} - \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} A(t,\rho)}{2\,W(t,\rho)} - \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{2\,W(t,\rho)^2} + \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{4\,W(t,\rho)^2} \\
& + \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} A(t,\rho)}{4\,W(t,\rho)} - \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} \\
& - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\,W(t,\rho)} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} - \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)}
\end{aligned}
\tag{86}
$$

Ricci components:

$$\Phi'_{00} = \Phi_{00} \tag{87}$$

$$\begin{aligned}\Phi'_{01} = &-\frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)^2 - z e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right) - \frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U\left(t,\rho\right)^2 \\ &- \frac{z e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} - \frac{z e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{z e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} \\ &- \frac{z e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{z e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{z e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)}\end{aligned} \tag{88}$$

$$\begin{aligned}\Phi'_{02} = &-\frac{1}{2}\, z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)^2 - z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right) - \frac{1}{2}\, z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U\left(t,\rho\right)^2 - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} \\ &- \frac{z^2 e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\ &- \frac{z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial t}U\left(t,\rho\right)}{W\left(t,\rho\right)} + \frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}U\left(t,\rho\right) - \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right)}{W\left(t,\rho\right)} \\ &- \frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}U\left(t,\rho\right) + \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)\frac{\partial}{\partial t}W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U\left(t,\rho\right)\frac{\partial}{\partial \rho}W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\ &- \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} - \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\ &- \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial t}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial \rho}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)}\end{aligned} \tag{89}$$

$$\begin{aligned}\Phi'_{10} = &-\frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)^2 - z e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right) - \frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U\left(t,\rho\right)^2 \\ &- \frac{z e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} - \frac{z e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{z e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} \\ &- \frac{z e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{z e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{z e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)}\end{aligned} \tag{90}$$

$$
\begin{aligned}
\Phi'_{11} = & -\frac{1}{2}\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - \frac{1}{2}\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 \\
& - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{8\,W(t,\rho)^2} - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{4\,W(t,\rho)^2} - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{8\,W(t,\rho)^2} \\
& - \frac{1}{4}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 + \frac{1}{4}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\,W(t,\rho)} \\
& - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)} + \frac{1}{4}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) - \frac{1}{4}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) \\
& + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)} - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)} - \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{16\,W(t,\rho)^2} + \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{16\,W(t,\rho)^2}
\end{aligned}
\tag{91}
$$

$$
\begin{aligned}
\Phi'_{12} = & -\frac{1}{2}\, z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - \frac{1}{2}\, z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 \\
& - \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{8\,W(t,\rho)^2} - \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{4\,W(t,\rho)^2} - \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{8\,W(t,\rho)^2} - \frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 \\
& + \frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 - \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} - \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\,W(t,\rho)} - \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)} \\
& + \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{W(t,\rho)} + z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) - \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{W(t,\rho)} \\
& - z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) + \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{W(t,\rho)} - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{W(t,\rho)} \\
& - \frac{3\, z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{8\,W(t,\rho)^2} + \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} A(t,\rho)}{4\,W(t,\rho)} + \frac{3\, z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{8\,W(t,\rho)^2} - \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} A(t,\rho)}{4\,W(t,\rho)} \\
& - \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} + \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)}
\end{aligned}
\tag{92}
$$

$$
\begin{aligned}
\Phi'_{20} = & -\frac{1}{2}\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right)^2 - z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right) \frac{\partial}{\partial \rho} U\left(t,\rho\right) - \frac{1}{2}\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right)^2 - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} \\
& - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial \rho} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial t} U\left(t,\rho\right)}{W\left(t,\rho\right)} + \frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U\left(t,\rho\right) + \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right) \frac{\partial}{\partial \rho} U\left(t,\rho\right)}{W\left(t,\rho\right)} \\
& - \frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U\left(t,\rho\right) + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right) \frac{\partial}{\partial t} W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right) \frac{\partial}{\partial \rho} W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& - \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} - \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} + \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\
& + \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial t} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right) \frac{\partial}{\partial \rho} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)}
\end{aligned}
\tag{93}
$$

$$
\begin{aligned}
\Phi'_{21} = & -\frac{1}{2}\, z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right)^2 - z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right) \frac{\partial}{\partial \rho} U\left(t,\rho\right) - \frac{1}{2}\, z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right)^2 \\
& - \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} - \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial \rho} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} - \frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right)^2 \\
& + \frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right)^2 - \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\
& - \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial t} U\left(t,\rho\right)}{W\left(t,\rho\right)} + z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U\left(t,\rho\right) + \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right) \frac{\partial}{\partial \rho} U\left(t,\rho\right)}{W\left(t,\rho\right)} \\
& - z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U\left(t,\rho\right) + \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right) \frac{\partial}{\partial t} W\left(t,\rho\right)}{W\left(t,\rho\right)} - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right) \frac{\partial}{\partial \rho} W\left(t,\rho\right)}{W\left(t,\rho\right)} \\
& - \frac{3\, z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} - \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{3\, z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} + \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\
& + \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial t} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right) \frac{\partial}{\partial \rho} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)}
\end{aligned}
\tag{94}
$$

$$
\begin{aligned}
\Phi'_{22} = & -\frac{1}{2}\, z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - \frac{1}{2}\, z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 - \frac{z^4 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{8\,W(t,\rho)^2} \\
& - \frac{z^4 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{4\,W(t,\rho)^2} - \frac{z^4 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{8\,W(t,\rho)^2} - z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 + z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 \\
& - \frac{z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} - \frac{z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\,W(t,\rho)} - \frac{z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)} + 2\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) \\
& - 2\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) + \frac{2\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{W(t,\rho)} - \frac{2\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{W(t,\rho)} - \frac{3\, z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{4\,W(t,\rho)^2} \\
& + \frac{3\, z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{4\,W(t,\rho)^2} - \frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 + e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - \frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 \\
& - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{2\,W(t,\rho)} + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{2\,W(t,\rho)} - \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{8\,W(t,\rho)^2} + \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{4\,W(t,\rho)^2} \\
& - \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{8\,W(t,\rho)^2} - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\,W(t,\rho)} - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)}
\end{aligned}
\tag{95}
$$

B Type transformations:

Spin coefficients:

$$\rho' = \frac{\sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} + \frac{\sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)} \tag{96}$$

$$\begin{aligned}\sigma' = {} & \frac{1}{2}\sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{2}\sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}z^2 e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{4\,W(t,\rho)} \\ & + \frac{i\sqrt{2}z^2 e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{4\,W(t,\rho)} - \frac{\sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} - \frac{\sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)}\end{aligned} \tag{97}$$

$$\kappa' = 0 \tag{98}$$

$$\mu' = -\frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,z^2 W(t,\rho)} + \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,z^2 W(t,\rho)} \tag{99}$$

$$\begin{aligned}\lambda' = {} & -\frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho)}{2\,z^2} + \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho)}{2\,z^2} + \frac{i\sqrt{2}e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{4\,z^2 W(t,\rho)} \\ & - \frac{i\sqrt{2}e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{4\,z^2 W(t,\rho)} + \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,z^2 W(t,\rho)} - \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,z^2 W(t,\rho)}\end{aligned} \tag{100}$$

$$\nu' = 0 \tag{101}$$

$$\alpha' = 0 \tag{102}$$

$$\beta' = 0 \tag{103}$$

$$\tau' = 0 \tag{104}$$

$$\pi' = 0 \tag{105}$$

$$\epsilon' = \frac{1}{4}\sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{4}\sqrt{2}z^2 e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}z^2 e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{8\,W(t,\rho)} + \frac{i\sqrt{2}z^2 e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{8\,W(t,\rho)} \tag{106}$$

$$\gamma' = -\frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho)}{4\,z^2} + \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho)}{4\,z^2} + \frac{i\sqrt{2}e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{8\,z^2 W(t,\rho)} - \frac{i\sqrt{2}e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{8\,z^2 W(t,\rho)} \tag{107}$$

Weyl components:

$$\begin{aligned}
\Psi_0' = {} & -z^4 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho)^2 - 2\,z^4 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho)\frac{\partial}{\partial \rho}U(t,\rho) - z^4 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho)^2 - \frac{3i\,z^4 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)\frac{\partial}{\partial t}U(t,\rho)}{2\,W(t,\rho)} \\
& - \frac{3i\,z^4 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)\frac{\partial}{\partial t}U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z^4 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}U(t,\rho) - z^4 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}U(t,\rho) - \frac{3i\,z^4 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)\frac{\partial}{\partial \rho}U(t,\rho)}{2\,W(t,\rho)} \\
& - \frac{3i\,z^4 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)\frac{\partial}{\partial \rho}U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z^4 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}U(t,\rho) + \frac{z^4 e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)^2}{4\,W(t,\rho)^2} - \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}A(t,\rho)}{4\,W(t,\rho)} \\
& - \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}A(t,\rho)}{2\,W(t,\rho)} + \frac{z^4 e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)\frac{\partial}{\partial \rho}A(t,\rho)}{2\,W(t,\rho)^2} + \frac{z^4 e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)^2}{4\,W(t,\rho)^2} - \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}A(t,\rho)}{4\,W(t,\rho)} \\
& + \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)^2} + \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z^4 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}W(t,\rho)}{4\,W(t,\rho)} + \frac{z^4 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}W(t,\rho)}{2\,W(t,\rho)} \\
& + \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)^2} + \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z^4 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}W(t,\rho)}{4\,W(t,\rho)}
\end{aligned} \tag{108}$$

$$\Psi_1' = 0 \tag{109}$$

$$\Psi_2' = \frac{1}{3}\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - \frac{1}{3}\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 + \frac{i\, z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\, W(t,\rho)} + \frac{1}{6}\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho)$$
$$- \frac{i\, z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\, W(t,\rho)} - \frac{1}{6}\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{3\, W(t,\rho)}$$
$$+ \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{3\, W(t,\rho)} + \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{12\, W(t,\rho)^2} - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{12\, W(t,\rho)^2} - \frac{i\, z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\, W(t,\rho)^2}$$
$$+ \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{12\, W(t,\rho)} + \frac{i\, z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\, W(t,\rho)^2} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{12\, W(t,\rho)} \tag{110}$$

$$\Psi_3' = 0 \tag{111}$$

$$\Psi_4' = \Psi_4 \tag{112}$$

Ricci components:

$$\Phi_{00}' = -\frac{1}{2}\, z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - \frac{1}{2}\, z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2$$
$$- \frac{z^4 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{8\, W(t,\rho)^2} - \frac{z^4 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{4\, W(t,\rho)^2} - \frac{z^4 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{8\, W(t,\rho)^2}$$
$$- \frac{z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\, W(t,\rho)} - \frac{z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\, W(t,\rho)} - \frac{z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\, W(t,\rho)} \tag{113}$$

$$\Phi_{01}' = 0 \tag{114}$$

$$\Phi'_{02} = \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)\,\frac{\partial}{\partial t}U(t,\rho)}{W(t,\rho)} + \frac{1}{2}\,z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}U(t,\rho) - \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)\,\frac{\partial}{\partial \rho}U(t,\rho)}{W(t,\rho)}$$
$$- \frac{1}{2}\,z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}U(t,\rho) + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho)\,\frac{\partial}{\partial t}W(t,\rho)}{2\,W(t,\rho)} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho)\,\frac{\partial}{\partial \rho}W(t,\rho)}{2\,W(t,\rho)}$$
$$- \frac{z^2 e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)^2}{4\,W(t,\rho)^2} + \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}A(t,\rho)}{4\,W(t,\rho)} + \frac{z^2 e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)^2}{4\,W(t,\rho)^2} - \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}A(t,\rho)}{4\,W(t,\rho)}$$
$$- \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)\,\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)^2} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}W(t,\rho)}{4\,W(t,\rho)} + \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)\,\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}W(t,\rho)}{4\,W(t,\rho)} \tag{115}$$

$$\Phi'_{10} = 0 \tag{116}$$

$$\Phi'_{11} = -\frac{1}{4}\,z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho)^2 + \frac{1}{4}\,z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho)^2 + \frac{1}{4}\,z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}U(t,\rho) - \frac{1}{4}\,z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}U(t,\rho)$$
$$+ \frac{z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho)\,\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho)\,\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)} - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)^2}{16\,W(t,\rho)^2} + \frac{z^2 e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)^2}{16\,W(t,\rho)^2} \tag{117}$$

$$\Phi'_{12} = 0 \tag{118}$$

$$\Phi'_{20} = -\frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)\,\frac{\partial}{\partial t}U(t,\rho)}{W(t,\rho)} + \frac{1}{2}\,z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}U(t,\rho) + \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)\,\frac{\partial}{\partial \rho}U(t,\rho)}{W(t,\rho)}$$
$$- \frac{1}{2}\,z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}U(t,\rho) + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho)\,\frac{\partial}{\partial t}W(t,\rho)}{2\,W(t,\rho)} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho)\,\frac{\partial}{\partial \rho}W(t,\rho)}{2\,W(t,\rho)}$$
$$- \frac{z^2 e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)^2}{4\,W(t,\rho)^2} - \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}A(t,\rho)}{4\,W(t,\rho)} + \frac{z^2 e^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)^2}{4\,W(t,\rho)^2} + \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}A(t,\rho)}{4\,W(t,\rho)}$$
$$+ \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)\,\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)^2} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}W(t,\rho)}{4\,W(t,\rho)} - \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)\,\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}W(t,\rho)}{4\,W(t,\rho)} \tag{119}$$

$$\Phi'_{21} = 0 \tag{120}$$

$$\Phi'_{22} = \Phi_{22} \tag{121}$$

C Type transformations:

Spin coefficients:

$$\rho' = -\frac{\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{2\,W(t,\rho)} + \frac{\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{2\,W(t,\rho)} + \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} + \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)} \tag{122}$$

$$\begin{aligned}\sigma' = &-\frac{1}{2}\sqrt{2}z^2e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{2}\sqrt{2}z^2e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}z^2e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{4\,W(t,\rho)} - \frac{i\sqrt{2}z^2e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{4\,W(t,\rho)}\\ &- \frac{\sqrt{2}z^2e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} + \frac{\sqrt{2}z^2e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)} + \frac{1}{2}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{2}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho)\\ &+ \frac{i\sqrt{2}e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{4\,W(t,\rho)} + \frac{i\sqrt{2}e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{4\,W(t,\rho)} - \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} - \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)}\end{aligned} \tag{123}$$

$$\begin{aligned}\kappa' = &-\sqrt{2}z^3e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \sqrt{2}z^3e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}z^3e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{2\,W(t,\rho)} - \frac{i\sqrt{2}z^3e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{2\,W(t,\rho)} + \sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho)\\ &+ \sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}ze^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{2\,W(t,\rho)} + \frac{i\sqrt{2}ze^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{2\,W(t,\rho)} - \frac{\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} - \frac{\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)}\end{aligned} \tag{124}$$

$$\begin{aligned}\mu' = &-\sqrt{2}z^2e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \sqrt{2}z^2e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}z^2e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{2\,W(t,\rho)} - \frac{i\sqrt{2}z^2e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{2\,W(t,\rho)}\\ &+ \frac{\sqrt{2}z^2e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} - \frac{\sqrt{2}z^2e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)} - \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} + \frac{\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)}\end{aligned} \tag{125}$$

$$\lambda' = \lambda \tag{126}$$

$$\nu' = 0 \tag{127}$$

$$\alpha' = -\frac{1}{4}\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{4}\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}ze^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{8\,W(t,\rho)} - \frac{i\sqrt{2}ze^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{8\,W(t,\rho)} \tag{128}$$

$$\begin{aligned}\beta' = &-\frac{1}{4}\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{4}\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}ze^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{8\,W(t,\rho)} \\ &- \frac{i\sqrt{2}ze^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{8\,W(t,\rho)} - \frac{\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial t}W(t,\rho)}{4\,W(t,\rho)} + \frac{\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}W(t,\rho)}{4\,W(t,\rho)}\end{aligned} \tag{129}$$

$$\tau' = -\frac{1}{2}\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{2}\sqrt{2}ze^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}ze^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{4\,W(t,\rho)} - \frac{i\sqrt{2}ze^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{4\,W(t,\rho)} \tag{130}$$

$$\pi' = 0 \tag{131}$$

$$\begin{aligned}\epsilon' = &-\frac{3}{4}\sqrt{2}z^2e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{3}{4}\sqrt{2}z^2e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{3i\sqrt{2}z^2e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{8\,W(t,\rho)} - \frac{3i\sqrt{2}z^2e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{8\,W(t,\rho)} \\ &+ \frac{1}{4}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial t}U(t,\rho) + \frac{1}{4}\sqrt{2}e^{(-k+U(t,\rho))}\frac{\partial}{\partial \rho}U(t,\rho) + \frac{i\sqrt{2}e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial t}A(t,\rho)}{8\,W(t,\rho)} + \frac{i\sqrt{2}e^{(-k+3\,U(t,\rho))}\frac{\partial}{\partial \rho}A(t,\rho)}{8\,W(t,\rho)}\end{aligned} \tag{132}$$

$$\gamma' = \gamma \tag{133}$$

Weyl components:

$$
\begin{aligned}
\Psi_0' = & -z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 + 2\,z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 + \frac{3i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} \\
& - \frac{3i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) + z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} U(t,\rho) - \frac{3i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} \\
& + \frac{3i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) + \frac{z^4 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{4\,W(t,\rho)^2} + \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} A(t,\rho)}{4\,W(t,\rho)} \\
& - \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} A(t,\rho)}{2\,W(t,\rho)} - \frac{z^4 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{2\,W(t,\rho)^2} + \frac{z^4 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{4\,W(t,\rho)^2} + \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} A(t,\rho)}{4\,W(t,\rho)} \\
& + 2\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - 2\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 - \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} \\
& + \frac{z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} - \frac{z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\,W(t,\rho)} + \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} - \frac{i\,z^4 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} \\
& + \frac{z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)} + \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{W(t,\rho)} + z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) - \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{W(t,\rho)} \\
& - z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) - \frac{2\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{W(t,\rho)} + \frac{2\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{W(t,\rho)} + \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{2\,W(t,\rho)^2} \\
& - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{2\,W(t,\rho)^2} - e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - 2\,e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 \\
& - \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{2\,W(t,\rho)^2} + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{2\,W(t,\rho)} + \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{2\,W(t,\rho)^2} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{2\,W(t,\rho)} \\
& - \frac{3i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} - \frac{3i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) - e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} U(t,\rho) \\
& - \frac{3i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} - \frac{3i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) + \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{4\,W(t,\rho)^2} \\
& - \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} A(t,\rho)}{4\,W(t,\rho)} - \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} A(t,\rho)}{2\,W(t,\rho)} + \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{2\,W(t,\rho)^2} + \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{4\,W(t,\rho)^2} \\
& - \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} A(t,\rho)}{4\,W(t,\rho)} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} \\
& + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\,W(t,\rho)} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)}
\end{aligned}
\tag{134}
$$

$$
\begin{aligned}
\Psi_1' = & -z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 + 2\,z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 + \frac{3i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} \\
& - \frac{3i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) + z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} U(t,\rho) \\
& - \frac{3i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} + \frac{3i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) \\
& + \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{4\,W(t,\rho)^2} + \frac{i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} A(t,\rho)}{4\,W(t,\rho)} - \frac{i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} A(t,\rho)}{2\,W(t,\rho)} - \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{2\,W(t,\rho)^2} \\
& + \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{4\,W(t,\rho)^2} + \frac{i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} A(t,\rho)}{4\,W(t,\rho)} + z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 \\
& - \frac{i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} - \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\,W(t,\rho)} \\
& + \frac{i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} - \frac{i\,z^3 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)} \\
& + \frac{3i\,z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} + \frac{1}{2}\,z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) - \frac{3i\,z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) \\
& - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{W(t,\rho)} + \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{W(t,\rho)} + \frac{z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{4\,W(t,\rho)^2} - \frac{z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{4\,W(t,\rho)^2} \\
& - \frac{3i\,z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} + \frac{3i\,z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)}
\end{aligned}
\tag{135}
$$

$$
\begin{aligned}
\Psi_2' = & -z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 + 2\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 + \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} \\
& - \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) + z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} U(t,\rho) \\
& - \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} + \frac{3i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{2}\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) \\
& + \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{4\,W(t,\rho)^2} + \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} A(t,\rho)}{4\,W(t,\rho)} - \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} A(t,\rho)}{2\,W(t,\rho)} - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{2\,W(t,\rho)^2} \\
& + \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{4\,W(t,\rho)^2} + \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} A(t,\rho)}{4\,W(t,\rho)} + \frac{1}{3}\,e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - \frac{1}{3}\,e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 \\
& - \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\,W(t,\rho)} \\
& + \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} - \frac{i\,z^2 e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)} \\
& + \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{2\,W(t,\rho)} + \frac{1}{6}\,e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) - \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{2\,W(t,\rho)} - \frac{1}{6}\,e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) \\
& - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{3\,W(t,\rho)} + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{3\,W(t,\rho)} + \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{12\,W(t,\rho)^2} - \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{12\,W(t,\rho)^2} \\
& - \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{12\,W(t,\rho)} + \frac{i\,e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{12\,W(t,\rho)}
\end{aligned}
\tag{136}
$$

$$
\begin{aligned}
\Psi_3' = & -ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)^2 + 2\,ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial t}U\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right) - ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial}{\partial \rho}U\left(t,\rho\right)^2 + \frac{3i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial t}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& - \frac{3i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial t}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{1}{2}\,ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}U\left(t,\rho\right) + ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}U\left(t,\rho\right) - \frac{3i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& + \frac{3i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial \rho}U\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{1}{2}\,ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}U\left(t,\rho\right) + \frac{ze^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} + \frac{i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\
& - \frac{i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}A\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{ze^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}A\left(t,\rho\right)}{2\,W\left(t,\rho\right)^2} + \frac{ze^{(-2\,k+6\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} + \frac{i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\
& - \frac{i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial t}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial t}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial t)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{\partial t\partial \rho}W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& + \frac{i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial t}A\left(t,\rho\right)\frac{\partial}{\partial \rho}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{i\,ze^{(-2\,k+4\,U(t,\rho))}\frac{\partial}{\partial \rho}A\left(t,\rho\right)\frac{\partial}{\partial \rho}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{ze^{(-2\,k+2\,U(t,\rho))}\frac{\partial^2}{(\partial \rho)^2}W\left(t,\rho\right)}{4\,W\left(t,\rho\right)}
\end{aligned}
\tag{137}
$$

$$
\Psi_4' = \Psi_4 \tag{138}
$$

Ricci components:

$$
\begin{aligned}
\Phi'_{00} = & -\frac{1}{2}\, z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 + z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - \frac{1}{2}\, z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 - \frac{z^4 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{8\,W(t,\rho)^2} \\
& + \frac{z^4 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{4\,W(t,\rho)^2} - \frac{z^4 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{8\,W(t,\rho)^2} - z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 + z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 \\
& - \frac{z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} + \frac{z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\,W(t,\rho)} - \frac{z^4 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)} + 2\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) \\
& - 2\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) + \frac{2\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{W(t,\rho)} - \frac{2\,z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{W(t,\rho)} - \frac{3\,z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{4\,W(t,\rho)^2} \\
& + \frac{3\,z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{4\,W(t,\rho)^2} - \frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 - e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - \frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 \\
& - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{2\,W(t,\rho)} + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{2\,W(t,\rho)} - \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{8\,W(t,\rho)^2} - \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{4\,W(t,\rho)^2} \\
& - \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{8\,W(t,\rho)^2} - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\,W(t,\rho)} - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)}
\end{aligned}
\tag{139}
$$

$$
\begin{aligned}
\Phi'_{01} = & -\frac{1}{2}\, z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 + z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho) - \frac{1}{2}\, z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 \\
& - \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{8\,W(t,\rho)^2} + \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial \rho} A(t,\rho)}{4\,W(t,\rho)^2} - \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{8\,W(t,\rho)^2} - \frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho)^2 \\
& + \frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho)^2 - \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} + \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W(t,\rho)}{2\,W(t,\rho)} - \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)} \\
& + \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} U(t,\rho)}{W(t,\rho)} + z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U(t,\rho) - \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} U(t,\rho)}{W(t,\rho)} \\
& - z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U(t,\rho) + \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{W(t,\rho)} - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{W(t,\rho)} \\
& - \frac{3\, z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho)^2}{8\,W(t,\rho)^2} + \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} A(t,\rho)}{4\,W(t,\rho)} + \frac{3\, z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho)^2}{8\,W(t,\rho)^2} - \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} A(t,\rho)}{4\,W(t,\rho)} \\
& - \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A(t,\rho) \frac{\partial}{\partial t} W(t,\rho)}{4\,W(t,\rho)^2} - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W(t,\rho)}{4\,W(t,\rho)} + \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A(t,\rho) \frac{\partial}{\partial \rho} W(t,\rho)}{4\,W(t,\rho)^2} + \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W(t,\rho)}{4\,W(t,\rho)}
\end{aligned}
\tag{140}
$$

$$
\begin{aligned}
\Phi'_{02} = & -\frac{1}{2}\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right)^2 + z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right) \frac{\partial}{\partial \rho} U\left(t,\rho\right) - \frac{1}{2}\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right)^2 - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} \\
& + \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial \rho} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial t} U\left(t,\rho\right)}{W\left(t,\rho\right)} + \frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U\left(t,\rho\right) - \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right) \frac{\partial}{\partial \rho} U\left(t,\rho\right)}{W\left(t,\rho\right)} \\
& - \frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U\left(t,\rho\right) + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right) \frac{\partial}{\partial t} W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right) \frac{\partial}{\partial \rho} W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& - \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} + \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} - \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\
& - \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial t} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right) \frac{\partial}{\partial \rho} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)}
\end{aligned}
\tag{141}
$$

$$
\begin{aligned}
\Phi'_{10} = & -\frac{1}{2}\, z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right)^2 + z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right) \frac{\partial}{\partial \rho} U\left(t,\rho\right) - \frac{1}{2}\, z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right)^2 \\
& - \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} + \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial \rho} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{z^3 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} - \frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right)^2 \\
& + \frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right)^2 - \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{z^3 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\
& - \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial t} U\left(t,\rho\right)}{W\left(t,\rho\right)} + z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U\left(t,\rho\right) + \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right) \frac{\partial}{\partial \rho} U\left(t,\rho\right)}{W\left(t,\rho\right)} \\
& - z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U\left(t,\rho\right) + \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right) \frac{\partial}{\partial t} W\left(t,\rho\right)}{W\left(t,\rho\right)} - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right) \frac{\partial}{\partial \rho} W\left(t,\rho\right)}{W\left(t,\rho\right)} \\
& - \frac{3\, z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} - \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{3\, z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} + \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\
& + \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial t} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{i\, z e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right) \frac{\partial}{\partial \rho} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)}
\end{aligned}
\tag{142}
$$

$$
\begin{aligned}
\Phi'_{11} = & -\frac{1}{2}\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right)^2 + z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right) \frac{\partial}{\partial \rho} U\left(t,\rho\right) - \frac{1}{2}\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right)^2 \\
& - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} + \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial \rho} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} \\
& - \frac{1}{4}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right)^2 + \frac{1}{4}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right)^2 - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{1}{4}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U\left(t,\rho\right) - \frac{1}{4}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U\left(t,\rho\right) \\
& + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right) \frac{\partial}{\partial t} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right) \frac{\partial}{\partial \rho} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right)^2}{16\,W\left(t,\rho\right)^2} + \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right)^2}{16\,W\left(t,\rho\right)^2}
\end{aligned}
\tag{143}
$$

$$
\begin{aligned}
\Phi'_{12} = & -\frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right)^2 + z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right) \frac{\partial}{\partial \rho} U\left(t,\rho\right) - \frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right)^2 \\
& - \frac{z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} + \frac{z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial \rho} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} \\
& - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)}
\end{aligned}
\tag{144}
$$

$$
\begin{aligned}
\Phi'_{20} = & -\frac{1}{2}\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right)^2 + z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right) \frac{\partial}{\partial \rho} U\left(t,\rho\right) - \frac{1}{2}\, z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right)^2 - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} \\
& + \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial \rho} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{z^2 e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& - \frac{z^2 e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial t} U\left(t,\rho\right)}{W\left(t,\rho\right)} + \frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} U\left(t,\rho\right) + \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right) \frac{\partial}{\partial \rho} U\left(t,\rho\right)}{W\left(t,\rho\right)} \\
& - \frac{1}{2}\, e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} U\left(t,\rho\right) + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right) \frac{\partial}{\partial t} W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right) \frac{\partial}{\partial \rho} W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} \\
& - \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} - \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right)^2}{4\,W\left(t,\rho\right)^2} + \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)} \\
& + \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial t} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} - \frac{i\, e^{(-2\,k+4\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right) \frac{\partial}{\partial \rho} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} + \frac{e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)}
\end{aligned}
\tag{145}
$$

$$
\begin{aligned}
\Phi'_{21} = & -\frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right)^2 + z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial t} U\left(t,\rho\right) \frac{\partial}{\partial \rho} U\left(t,\rho\right) - \frac{1}{2}\, z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial}{\partial \rho} U\left(t,\rho\right)^2 \\
& - \frac{z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} + \frac{z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial t} A\left(t,\rho\right) \frac{\partial}{\partial \rho} A\left(t,\rho\right)}{4\,W\left(t,\rho\right)^2} - \frac{z e^{(-2\,k+6\,U(t,\rho))} \frac{\partial}{\partial \rho} A\left(t,\rho\right)^2}{8\,W\left(t,\rho\right)^2} \\
& - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial t)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)} + \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{\partial t \partial \rho} W\left(t,\rho\right)}{2\,W\left(t,\rho\right)} - \frac{z e^{(-2\,k+2\,U(t,\rho))} \frac{\partial^2}{(\partial \rho)^2} W\left(t,\rho\right)}{4\,W\left(t,\rho\right)}
\end{aligned}
\tag{146}
$$

$$
\Phi'_{22} = \Phi_{22} \tag{147}
$$

References

[1] H. Stephani, D. Kramer, M. MacCallum, C. Hoenselaers, and E. Herlt, Exact Solutions of Einstein's Field Equations. Cambridge Monographs on Mathematical Physics, Cambridge University Press, 2009.

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